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ELECTRONIC DEVICE, ELECTRONIC STRUCTURE AND METHOD OF MANUFACTURING THE SAME

Abstract

An electronic device, an electronic structure and a manufacturing method are provided. The electronic device includes a substrate, a conductive structure and at least one external connector. The conductive structure is formed on the substrate and includes a test pad configured to be contacted by a probe during a testing process. The external connector is electrically connected to the conductive structure and is exposed from a surface of the electronic device for an external electrical connection. A vertical projection of the at least one external connector overlaps a vertical projection of the test pad.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of U.S. Non-Provisional application Ser. No. 17/954,752 filed Sep. 28, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an electronic device, an electronic structure and a method of manufacturing the same, and more particularly, to an electronic device including a test pad, and a method of manufacturing the same.

DISCUSSION OF THE BACKGROUND

[0003] Semiconductor structures are used in a variety of electronic applications, and the dimensions of semiconductor structures are continuously being scaled down to meet the current application requirements. However, a variety of issues arise during the scaling-down process and impact the final electrical characteristics, quality, cost and yield.

[0004] This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed herein constitutes prior art with respect to the present disclosure, and no part of this Discussion of the Background may be used as an admission that any part of this application constitutes prior art with respect to the present disclosure.

SUMMARY

[0005] One aspect of the present disclosure provides an electronic device including a substrate, a conductive structure and at least one external connector. The conductive structure is disposed on the substrate and includes a test pad configured to be contacted by a probe during a testing process. The external connector is electrically connected to the conductive structure and is exposed from a surface of the electronic device for an external electrical connection. A vertical projection of the at least one external connector overlaps a vertical projection of the test pad.

[0006] Another aspect of the present disclosure provides an electronic structure including a first substrate, a first conductive structure and a second conductive structure. The first conductive structure is disposed over the first substrate, and includes a first test pad configured to be contacted by a probe during a testing process. The second conductive structure is disposed under the first substrate, and includes a second test pad configured to be contacted by a probe during a testing process. An electrical path between the first conductive structure and the second conductive structure is located between the first test pad and the second test pad.

[0007] Another aspect of the present disclosure provides a manufacturing method. The manufacturing method includes forming a first conductive via in a first substrate and a first conductive structure on the first substrate, wherein the first conductive via is electrically connected to the first conductive structure, the first conductive structure defines a first opening to expose a first test pad thereof, and the first conductive via is disposed under the first test pad; testing the first test pad; and thinning the first substrate to expose the first conductive via.

[0008] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent

constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the Figures, where like reference numbers refer to similar elements throughout the Figures, and:

[0010] FIG. 1 is a schematic cross-sectional view of an electronic structure in accordance with some embodiments of the present disclosure.

[0011] FIG. 2 is an enlarged cross-sectional view of the first portion of the electronic structure of FIG. 1.

[0012] FIG. 3 is an enlarged cross-sectional view of the second portion of the electronic structure of FIG. 1.

[0013] FIG. 4 is an enlarged cross-sectional view of the third portion of the electronic structure of FIG. 1.

[0014] FIG. 5 is an enlarged cross-sectional view of the fourth portion of the electronic structure of FIG. 1.

[0015] FIG. 6 is a partially enlarged top view of the electronic structure of FIG. 1.

[0016] FIG. 7 is a schematic cross-sectional view of an electronic structure in accordance with some embodiments of the present disclosure.

[0017] FIG. 8 is an enlarged cross-sectional view of the first portion of the electronic structure of FIG. 7.

[0018] FIG. 9 is an enlarged cross-sectional view of the second portion of the electronic structure of FIG. 7.

[0019] FIG. 10 is an enlarged cross-sectional view of the third portion of the electronic structure of FIG. 7.

[0020] FIG. 11 is an enlarged cross-sectional view of the fourth portion of the electronic structure of FIG. 7.

[0021] FIG. 12 is a schematic cross-sectional view of an electronic structure in accordance with some embodiments of the present disclosure.

[0022] FIG. 13 is a partially enlarged top view of the electronic structure of FIG. 12.

[0023] FIG. 14 is a schematic cross-sectional view of an electronic structure in accordance with some embodiments of the present disclosure.

[0024] FIG. 15 is a partially enlarged top view of the electronic structure of FIG. 14.

[0025] FIG. 16 to FIG. 25 illustrate various stages of a method of manufacturing an electronic structure, in accordance with some embodiments of the present disclosure.

[0026] FIG. 26 is a flowchart of a method of manufacturing an electronic structure, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] Embodiments, or examples, of the disclosure illustrated in the drawings are now described using specific language. It shall be understood that no limitation of the scope of the disclosure is hereby intended. Any alteration or modification of the described embodiments, and any further applications of principles described in this document, are to be considered as normally occurring to one of ordinary skill in the art to which the disclosure relates. Reference numerals may be repeated throughout the embodiments, but this does not necessarily mean that feature(s) of one embodiment apply to another embodiment, even if they share the same reference numeral.

[0028] It shall be understood that, although the terms first, second, third, etc., may be used herein

to describe various elements, components, regions, layers or sections, these elements, components, regions, layers or sections are not limited by these terms. Rather, these terms are merely used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present inventive concept.

[0029] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limited to the present inventive concept. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It shall be further understood that the terms “comprises” and “comprising,” when used in this specification, point out the presence of stated features, integers, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or groups thereof.

[0030] FIG. 1 is a schematic cross-sectional view of an electronic structure 5 in accordance with some embodiments of the present disclosure. In some embodiments, the electronic structure 1 may be a semiconductor structure or a semiconductor device that includes a plurality of electronic devices stacked on one another. Thus, the electronic structure 1 may be a stacked structure that includes a plurality stacked memory devices (e.g., dynamic random access memories (DRAMs)). For example, the electronic structure 1 may be a high bandwidth memory (HBM). In some embodiments, the electronic structure 1 may include a first portion (e.g., a first electronic device 1), a second portion (e.g., a second electronic device 2), a third portion (e.g., a third electronic device 3) and a fourth portion (e.g., a fourth electronic device 4). The third portion (e.g., the third electronic device 3) is stacked on and connected to the fourth portion (e.g., the fourth electronic device 4). The second portion (e.g., a second electronic device 2) is stacked on and connected to the third portion (e.g., the third electronic device 3). The first portion (e.g., the first electronic device 1) is stacked on and connected to the second portion (e.g., a second electronic device 2).

[0031] FIG. 2 is an enlarged cross-sectional view of the first portion of the electronic structure 5 of FIG. 1. The first portion may be a first electronic device 1. The first electronic device 1 may be or include a portion of an integrated circuit (IC) chip that includes various passive and active microelectronic devices, such as resistors, capacitors, inductors, diodes, p-type field-effect transistors (pFETs), n-type field-effect transistors (nFETs), metal-oxide semiconductor field-effect transistors (MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJTs), laterally-diffused MOS (LDMOS) transistors, high-voltage transistors, high-frequency transistors, fin field-effect transistors (FinFETs), other suitable IC components, or combinations thereof.

[0032] The first electronic device 1 may have a first surface 11 (e.g., a top surface) and a second surface 12 (e.g., a bottom surface) opposite to the first surface 11. The first electronic device 1 may include a first substrate 10, a capacitor cell 13, a first conductive structure 14, a first bottom passivation layer 16 and at least one first external connector (e.g., a first conductive via 15).

[0033] In some embodiments, the first substrate 10 (e.g., a semiconductor substrate) may have a first surface 101 (e.g., a top surface) and a second surface 102 (e.g., a bottom surface) opposite to the first surface 101. The first substrate 10 may include, for example, silicon (Si), doped silicon, germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), gallium (Ga), gallium arsenide (GaAs), indium (In), indium arsenide (InAs), indium phosphide (InP) or other IV-IV, III-V or II-VI semiconductor materials. In some other embodiments, the first substrate 10 may include a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate.

[0034] Depending on the IC fabrication stage, the first substrate 10 may include various material

layers (e.g., dielectric layers, semiconductor layers, and/or conductive layers) configured to form IC features (e.g., doped regions, isolation features, gate features, source/drain features, interconnect features, other features, or combinations thereof).

[0035] The capacitor cell **13** may be disposed on or disposed over the first surface **101** of the first substrate **10**. In some embodiments, the capacitor cell **13** may be embedded in the first substrate **10**.

[0036] The first conductive structure **14** may be disposed on or disposed over the first surface **101** of the first substrate **10**, may have a first surface **141** (e.g., a top surface). The first surface **141** of the first conductive structure **14** may be the first surface **11** of the first electronic device **1**. The first conductive structure **14** may include a plurality of patterned metal layers **142**, a plurality of inner vias **143**, a first test pad **144**, at least one inner via **145** and a dielectric structure **146**. The dielectric structure **146** may include one or more dielectric layers. The patterned metal layers **142**, the inner vias **143**, the first test pad **144**, the inner via **145** and the capacitor cell **13** may be embedded in or may be covered by the dielectric structure **146**.

[0037] The patterned metal layers **142** may be patterned circuit layers **142**, and may be electrically connected to each other by the inner vias **143**. The patterned metal layers **142** may be a back-end-of-line (BEOL) or a front-end-of-line (FEOL). A material of the patterned metal layers **142** and the inner vias **143** may include copper (Cu). The first test pad **144** may be electrically connected to the patterned metal layers **142** through the inner via **145**. A material of the first test pad **144** may include aluminum (Al), a material of the inner via **145** may include tungsten (W).

[0038] The first test pad **144** may be configured to be contacted by a probe during a testing process. The first test pad **144** may include a first portion **144a** and a second portion **144b**. The dielectric structure **146** may define a first opening **147** to expose the first portion **144a** of the first test pad **144**. The first portion **144a** may be configured to be contacted by the probe during the testing process. Thus, the first portion of **144a** the first test pad **144** may have a probe mark **148** thereon after the testing process. The probe mark **148** may be a recess portion recessed from a top surface of the first test pad **144**. In addition, the second portion **144b** may be covered by the dielectric structure **146**. That is, the size of the first opening **147** may be less than the size of the first test pad **144**.

[0039] The first bottom passivation layer **16** may be disposed on the second surface **102** of the first substrate **10** and may surround the first external connector (e.g., a first conductive via **15**). The first bottom passivation layer **16** may have a second surface **162** (e.g., a bottom surface). The second surface **162** of the first bottom passivation layer **16** may be the second surface **12** of the first electronic device **1**. A material of the first bottom passivation layer **16** may include oxide material or nitride material, such as silicon nitride (Si.sub.3N.sub.4, or SiN), silicon dioxide (SiO.sub.2), silicon oxynitride (N.sub.2OSi.sub.2), silicon nitride oxide (SiON), tantalum pentoxide (Ta.sub.2O.sub.5), aluminum oxide (Al.sub.2O.sub.3), strontium bismuth tantalum oxide (SrBi.sub.2Ta.sub.2O.sub.9, SBT), barium strontium titanate oxide (BaSrTiO.sub.3, BST), or a combination thereof.

[0040] The first external connector may include a first conductive via **15**. The first conductive via **15** may be electrically connected to the patterned metal layers **142** of the first conductive structure **14**, and may be exposed from the second surface **12** of the first electronic device **1** for an external electrical connection. A material of the first conductive via **15** may be copper. The first conductive via **15** may extend through the first substrate **10**, and may be disposed right under the patterned metal layers **142** of the first conductive structure **14**.

[0041] As shown in FIG. **1**, a length of the first conductive via **15** may be greater than a thickness of the first substrate **10**. Thus, the first conductive via **15** may extend beyond the second surface **102** of the first substrate **10**. In addition, the first conductive via **15** may further extend through the first bottom passivation layer **16**. A second surface **152** (e.g., a bottom surface) of the first conductive via **15** may be substantially aligned with the second surface **162** of the first bottom passivation layer **16**.

[0042] FIG. 3 is an enlarged cross-sectional view of the second portion of the electronic structure 5 of FIG. 1. The second portion may be a second electronic device 2. The second electronic device 2 may be similar to the first electronic device 1 of FIG. 2.

[0043] The second electronic device 2 may have a first surface 21 (e.g., a top surface) and a second surface 22 (e.g., a bottom surface) opposite to the first surface 21. The second electronic device 2 may include a second substrate 20, a capacitor cell 23, a second conductive structure 24, a second bottom passivation layer 26, at least one second external connector (e.g., a second conductive via 25 or a second connection via 28) and a second top passivation layer 27.

[0044] In some embodiments, the second substrate 20 may be same as or similar to the first substrate 10 of FIG. 2. The second substrate 20 may have a first surface 201 (e.g., a top surface) and a second surface 202 (e.g., a bottom surface) opposite to the first surface 201. The capacitor cell 23 may be disposed on or disposed over the first surface 201 of the second substrate 20. In some embodiments, the capacitor cell 23 may be embedded in the second substrate 20.

[0045] The second conductive structure 24 may be same as or similar to the first conductive structure 14 of FIG. 2. The second conductive structure 24 may be disposed on or disposed over the first surface 201 of the second substrate 20, may have a first surface 241 (e.g., a top surface).

[0046] The second conductive structure 24 may include a plurality of patterned metal layers 242, a plurality of inner vias 243, a second test pad 244, at least one inner via 245 and a dielectric structure 246. The dielectric structure 246 may include one or more dielectric layers. The patterned metal layers 242, the inner vias 243, the second test pad 244, the inner via 245 and the capacitor cell 23 may be embedded in or may be covered by the dielectric structure 246.

[0047] The patterned metal layers 242 may be patterned circuit layers 242, and may be electrically connected to each other by the inner vias 243. The second test pad 244 may be electrically connected to the patterned metal layers 242 through the inner via 245.

[0048] The second test pad 244 may be configured to be contacted by a probe during a testing process. The second test pad 244 may include a first portion 244a and a second portion 244b. The dielectric structure 246 may define a second opening 247 to expose the first portion 244a of the second test pad 244. The first portion 244a may be configured to be contacted by the probe during the testing process. Thus, the first portion 244a of the second test pad 244 may have a probe mark 248 thereon after the testing process. In addition, the second portion 244b may be covered by a portion 2461 of the dielectric structure 246. That is, the size of the second opening 247 may be less than the size of the second test pad 244.

[0049] The second top passivation layer 27 may be disposed on or disposed over the first surface 241 of the second conductive structure 24, and may surround the second external connector (e.g., a second connection via 28). As shown in FIG. 3, the second top passivation layer 27 may cover the first surface 241 of the second conductive structure 24, and may extend into the second opening 247 and the probe mark 248. The second top passivation layer 27 may have a first surface 271 (e.g., a top surface). The first surface 271 of the second top passivation layer 27 may be the first surface 21 of the second electronic device 2. A material of the second top passivation layer 27 may include oxide material or nitride material, such as silicon nitride (Si_{3}N_4 , or SiN), silicon dioxide (SiO_2), silicon oxynitride ($\text{N}_{2.0}\text{Si}_{2.0}$), silicon nitride oxide (SiON), tantalum pentoxide ($\text{Ta}_{2.0}\text{O}_{5.0}$), aluminum oxide (Al_2O_3), strontium bismuth tantalum oxide ($\text{SrBi}_{2.0}\text{Ta}_{2.0}\text{O}_{9.0}$, SBT), barium strontium titanate oxide ($\text{BaSrTiO}_{3.0}$, BST), or a combination thereof.

[0050] The second external connector may include a second connection via 28. The second connection via 28 may be electrically connected and physically connected to the second portion 244b of the second test pad 244, and may be exposed from the first surface 21 of the second electronic device 2 for an external electrical connection. A material of the second connection via 28 may be copper. The second connection via 28 may extend through the second top passivation layer 27 and the portion 2461 of the dielectric structure 246 of the second conductive structure 24 on the

second test pad **244**, and may be disposed right above the second test pad **244**. A first surface **281** (e.g., a top surface) of the second connection via **28** may be substantially aligned with the first surface **271** of the second top passivation layer **27**. Thus, the second connection via **28** may be exposed from the first surface **21** of the second electronic device **2** for external electrical connection. The second connection via **28** may be also referred to as a contact via. In addition, the second test pad **244** may be disposed between the second conductive via **25** and the second connection via **28** in a vertical direction.

[0051] The second bottom passivation layer **26** may be disposed on or disposed over the second surface **202** of the second substrate **20** and may surround the second external connector (e.g., a second conductive via **25**). The second bottom passivation layer **26** may have a second surface **262** (e.g., a bottom surface). The second surface **262** of the second bottom passivation layer **26** may be the second surface **22** of the second electronic device **2**.

[0052] The second external connector may further include a second conductive via **25**. The second conductive via **25** may be electrically connected to the patterned metal layers **242** of the second conductive structure **24**, and may be exposed from the second surface **22** of the second electronic device **2** for an external electrical connection. The second conductive via **25** may extend through the second substrate **20**, and may be disposed right under the patterned metal layers **242** of the second conductive structure **24**.

[0053] As shown in FIG. **3**, a length of the second conductive via **25** may be greater than a thickness of the second substrate **20**. Thus, the second conductive via **25** may extend beyond the second surface **202** of the second substrate **20**. In addition, the second conductive via **25** may further extend through the second bottom passivation layer **26**. A second surface **252** (e.g., a bottom surface) of the second conductive via **25** may be substantially aligned with the second surface **262** of the second bottom passivation layer **26**. In addition, a width **W21** of the second connection via **28** may be less than a width **W22** of the second conductive via **25**. The width **W21** of the second connection via **28** may be less than a width **W12** of the first conductive via **15** of FIG. **2**.

[0054] FIG. **4** is an enlarged cross-sectional view of the third portion of the electronic structure **5** of FIG. **1**. The third portion may be a third electronic device **3**. The third electronic device **3** may be same as or similar to the second electronic device **2** of FIG. **3**.

[0055] The third electronic device **3** may have a first surface **31** (e.g., a top surface) and a second surface **32** (e.g., a bottom surface) opposite to the first surface **31**. The third electronic device **3** may include a third substrate **30**, a capacitor cell **33**, a third conductive structure **34**, a third bottom passivation layer **36**, at least one third external connector (e.g., a third conductive via **35** or a third connection via **38**) and a third top passivation layer **37**.

[0056] In some embodiments, the third substrate **30** may be same as or similar to the second substrate **20** of FIG. **3**. The third substrate **30** may have a first surface **301** (e.g., a top surface) and a second surface **302** (e.g., a bottom surface) opposite to the first surface **301**. The capacitor cell **33** may be disposed on or disposed over the first surface **301** of the third substrate **30**.

[0057] The third conductive structure **34** may be same as or similar to the second conductive structure **24** of FIG. **3**. The third conductive structure **34** may be disposed on or disposed over the first surface **301** of the third substrate **30**, may have a first surface **341** (e.g., a top surface).

[0058] The third conductive structure **34** may include a plurality of patterned metal layers **342**, a plurality of inner vias **343**, a third test pad **344**, at least one inner via **345** and a dielectric structure **346**. The dielectric structure **346** may include one or more dielectric layers. The patterned metal layers **342**, the inner vias **343**, the third test pad **344**, the inner via **345** and the capacitor cell **33** may be embedded in or may be covered by the dielectric structure **346**.

[0059] The patterned metal layers **342** may be patterned circuit layers

[0060] **342**, and may be electrically connected to each other by the inner vias **343**. The third test pad **344** may be electrically connected to the patterned metal layers **342** through the inner via **345**.

[0061] The third test pad **344** may include a first portion **344a** and a second portion **344b**. The

dielectric structure **346** may define a third opening **347** to expose the first portion **344a** of the third test pad **344**. The first portion **344a** may be configured to be contacted by the probe during the testing process. Thus, the first portion **344a** of the third test pad **344** may have a probe mark **348** thereon after the testing process. In addition, the second portion **344b** may be covered by the dielectric structure **346**.

[0062] The third top passivation layer **37** may be disposed on or disposed over the first surface **341** of the third conductive structure **34**, and may surround the third external connector (e.g., a third connection via **38**). As shown in FIG. **4**, the third top passivation layer **37** may cover the first surface **341** of the third conductive structure **34**, and may extend into the third opening **347** and the probe mark **348**. The third top passivation layer **37** may have a first surface **371** (e.g., a top surface). The first surface **371** of the third top passivation layer **37** may be the first surface **31** of the third electronic device **3**.

[0063] The third external connector may include a third connection via **38**. The third connection via **38** may be electrically connected and physically connected to the second portion **344b** of the third test pad **344**, and may be exposed from the first surface **31** of the third electronic device **3** for an external electrical connection. The third connection via **38** may extend through the third top passivation layer **37** and a portion of the dielectric structure **346** of the third conductive structure **34** on the third test pad **344**, and may be disposed right above the third test pad **344**.

[0064] A first surface **381** (e.g., a top surface) of the third connection via **38** may be substantially aligned with the first surface **371** of the third top passivation layer **37**. Thus, the third connection via **38** may be exposed from the first surface **31** of the third electronic device **3** for external electrical connection.

[0065] The third bottom passivation layer **36** may be disposed on or disposed over the second surface **302** of the third substrate **30** and may surround the third external connector (e.g., a third conductive via **35**). The third bottom passivation layer **36** may have a second surface **362** (e.g., a bottom surface). The second surface **362** of the third bottom passivation layer **36** may be the second surface **32** of the third electronic device **3**.

[0066] The third external connector may further include a third conductive via **35**. The third conductive via **35** may be electrically connected to the patterned metal layers **342** of the third conductive structure **34**, and may be exposed from the second surface **32** of the third electronic device **3** for an external electrical connection. The third conductive via **35** may extend through the third substrate **30**, and may be disposed right under the patterned metal layers **342** of the third conductive structure **34**.

[0067] The third conductive via **35** may extend beyond the second surface **302** of the third substrate **30**. In addition, the third conductive via **35** may further extend through the third bottom passivation layer **36**. A second surface **352** (e.g., a bottom surface) of the third conductive via **35** may be substantially aligned with the second surface **362** of the third bottom passivation layer **36**. In addition, a width of the third connection via **38** may be less than a width of the third conductive via **35**. The width of the third connection via **38** may be less than a width **W22** of the second conductive via **25** of FIG. **3**.

[0068] FIG. **5** is an enlarged cross-sectional view of the fourth portion of the electronic structure **5** of FIG. **1**. The fourth portion may be a fourth electronic device **4**. The fourth electronic device **4** may be same as or similar to the third electronic device **3** of FIG. **4**.

[0069] The fourth electronic device **4** may have a first surface **41** (e.g., a top surface) and a second surface **42** (e.g., a bottom surface) opposite to the first surface **41**. The fourth electronic device **4** may include a fourth substrate **40**, a capacitor cell **43**, a fourth conductive structure **44**, at least one fourth external connector (e.g., a fourth connection via **48**) and a fourth top passivation layer **47**.

[0070] In some embodiments, the fourth substrate **40** may be same as or similar to the third substrate **30** of FIG. **4**. The fourth substrate **40** may have a first surface **401** (e.g., a top surface) and a second surface **402** (e.g., a bottom surface) opposite to the first surface **401**. The capacitor cell **43**

may be disposed on or disposed over the first surface **401** of the fourth substrate **40**.

[0071] The fourth conductive structure **44** may be same as or similar to the third conductive structure **34** of FIG. **4**. The fourth conductive structure **44** may be disposed on or disposed over the first surface **401** of the fourth substrate **40**, may have a first surface **441** (e.g., a top surface).

[0072] The fourth conductive structure **44** may include a plurality of patterned metal layers **442**, a plurality of inner vias **443**, a fourth test pad **444**, at least one inner via **445** and a dielectric structure **446**. The dielectric structure **446** may include one or more dielectric layers. The patterned metal layers **442**, the inner vias **443**, the fourth test pad **444**, the inner via **445** and the capacitor cell **43** may be embedded in or may be covered by the dielectric structure **446**.

[0073] The patterned metal layers **442** may be patterned circuit layers **442**, and may be electrically connected to each other by the inner vias **443**. The fourth test pad **444** may be electrically connected to the patterned metal layers **442** through the inner via **445**.

[0074] The fourth test pad **444** may include a first portion **444a** and a second portion **444b**. The dielectric structure **446** may define a fourth opening **447** to expose the first portion **444a** of the fourth test pad **444**.

[0075] The first portion **444a** may be configured to be contacted by the probe during the testing process. Thus, the first portion **444a** of the fourth test pad **444** may have a probe mark **448** thereon after the testing process. In addition, the second portion **444b** may be covered by the dielectric structure **446**.

[0076] The fourth top passivation layer **47** may be disposed on or disposed over the first surface **441** of the fourth conductive structure **44**, and may surround the fourth external connector (e.g., a fourth connection via **48**). As shown in FIG. **5**, the fourth top passivation layer **47** may cover the first surface **441** of the fourth conductive structure **44**, and may extend into the fourth opening **447** and the probe mark **448**. The fourth top passivation layer **47** may have a first surface **471** (e.g., a top surface). The first surface **471** of the fourth top passivation layer **47** may be the first surface **41** of the fourth electronic device **4**.

[0077] The fourth external connector may include a fourth connection via **48**. The fourth connection via **48** may be electrically connected and physically connected to the second portion **444b** of the fourth test pad **444**, and may be exposed from the first surface **41** of the fourth electronic device **4** for an external electrical connection. The fourth connection via **48** may extend through the fourth top passivation layer **47** and a portion of the dielectric structure **446** of the fourth conductive structure **44** on the fourth test pad **444**, and may be disposed right above the fourth test pad **444**.

[0078] A first surface **481** (e.g., a top surface) of the fourth connection via **48** may be substantially aligned with the first surface **471** of the fourth top passivation layer **47**. Thus, the fourth connection via **48** may be exposed from the first surface **41** of the fourth electronic device **4** for external electrical connection.

[0079] As shown in FIG. **1**, the second conductive structure **24** may be disposed under the first substrate **10**, and the first conductive structure **14** of the first electronic device **1** may be electrically connected to the second conductive structure **24** of the second electronic device **2** through the first conductive via **15** and the second connection via **28** so as to form a first vertical electrical path **71**. Thus, a first electrical path between the first conductive structure **14** of the first electronic device **1** and the second conductive structure **24** of the second electronic device **2** may include the first vertical electrical path **71**. The first electrical path (including the first vertical electrical path **71**) may be located between the first test pad **144** and the second test pad **244**. For example, the first electrical path (including the first vertical electrical path **71**) is within a vertical projection of the first test pad **144**. Thus, a projection of the first vertical electrical path **71** on the second test pad **244** is within a projection of the first test pad **144** on the second test pad **244**. Further, the first vertical electrical path **71** may pass through the first substrate **10**. In addition, the first electrical path may pass through the second portion **244b** of the second test pad **244**. Thus, the second

portion **244b** of the second test pad **244** may be a portion of the first electrical path between the first conductive structure **14** of the first electronic device **1** and the second conductive structure **24** of the second electronic device **2**.

[0080] Similarly, the third conductive structure **34** may be disposed under the second substrate **20**, and the second conductive structure **24** of the second electronic device **2** may be electrically connected to the third conductive structure **34** of the third electronic device **3** through the second conductive via **25** and the third connection via **38** so as to form a second vertical electrical path **72**. Thus, a second electrical path between the second conductive structure **24** of the second electronic device **2** and the third conductive structure **34** of the third electronic device **3** may include the second vertical electrical path **72**. The second electrical path (including the second vertical electrical path **72**) may be located between the second test pad **244** and the third test pad **344**. For example, the second electrical path (including the second vertical electrical path **72**) is within a vertical projection of the second test pad **244**. Thus, a projection of the second vertical electrical path **72** on the third test pad **344** is within a projection of the second test pad **244** on the third test pad **344**. Further, the second vertical electrical path **72** may pass through the second substrate **20**. In addition, the second electrical path may pass through the second portion **344b** of the third test pad **344**. Thus, the second portion **344b** of the third test pad **344** may be a portion of the second electrical path between the second conductive structure **24** of the second electronic device **2** and the third conductive structure **34** of the third electronic device **3**.

[0081] Similarly, the fourth conductive structure **44** may be disposed under the third substrate **30**, and the third conductive structure **34** of the third electronic device **3** may be electrically connected to the fourth conductive structure **44** of the fourth electronic device **4** through the third conductive via **35** and the fourth connection via **48** so as to form a third vertical electrical path **73**. Thus, a third electrical path between the third conductive structure **34** of the third electronic device **3** and the fourth conductive structure **44** of the fourth electronic device **4** may include the third vertical electrical path **73**. The third electrical path (including the third vertical electrical path **73**) may be located between the third test pad **344** and the fourth test pad **444**. For example, the third electrical path (including the third vertical electrical path **73**) is within a vertical projection of the third test pad **344**. Thus, a projection of the third vertical electrical path **73** on the fourth test pad **444** is within a projection of the third test pad **344** on the fourth test pad **444**. Further, the third vertical electrical path **73** may pass through the third substrate **30**. In addition, the third electrical path may pass through the second portion **444b** of the fourth test pad **444**. Thus, the second portion **444b** of the fourth test pad **444** may be a portion of the third electrical path between the third conductive structure **34** of the third electronic device **3** and the fourth conductive structure **44** of the fourth electronic device **4**.

[0082] FIG. **6** is a partially enlarged top view of the electronic structure **5** of FIG. **1**. As shown in FIG. **1** and FIG. **6**, a vertical projection of the external connector (including, for example, the first conductive via **15** and the second connection via **28**) may overlap or may be disposed within a vertical projection of the first test pad **144**. In addition, the vertical projection of the external connector (including, for example, the first conductive via **15**) may overlap a vertical projection of the first opening **147**. The vertical projection of the external connector (including, for example, the second connection via **28**) may be located outside the vertical projection of the first opening **147**.

[0083] In the embodiment illustrated in FIG. **1** to FIG. **6**, the first electronic device **1**, the second electronic device **2**, the third electronic device **3** and the fourth electronic device **4** may be bonded directly to one another by hybrid bonding. Thus, a height of the electronic structure **5** may be reduced. Further, the bonding solder and underfill are not used so as to lower the manufacturing cost of the electronic structure **5** and avoid the high resistance issue. In addition, the external connector (including, for example, the first conductive via **15** and the second connection via **28**) may be located right under the first test pad **144**, thus, a vertical space occupied by the external connector (including, for example, the first conductive via **15** and the second connection via **28**)

and the first test pad **144** may be reduced, which may be an efficient space design. In addition, the test pad **144**, **244**, **344**, **444** may be used for probing and may be a portion of the electrical path, thus, the flexibility of layout design is improved.

[0084] FIG. 7 is a schematic cross-sectional view of an electronic structure **5a** in accordance with some embodiments of the present disclosure. The electronic structure **5a** may be similar to the electronic structure **5** of FIG. 1, and the differences are described as follows.

[0085] The first conductive via **15** and the second connection via **28** may contact each other or may be fused together to form a first interconnection pillar **54**. The first interconnection pillar **54** may be a monolithic structure, and there may be no interface between the first conductive via **15** and the second connection via **28**. The first interconnection pillar **54** may electrically connecting the first conductive structure **14** and the second conductive structure **24**, and may form the first electrical path (including the first vertical electrical path **71**). In addition, the first bottom passivation layer **16** and the second top passivation layer **27** may contact each other or may be fused together to form a first bonding layer **51**. The first bonding layer **51** may be a monolithic structure, and there may be no interface between the first bottom passivation layer **16** and the second top passivation layer **27**. The first bonding layer **51** may be used for bonding the first substrate **10** and the second conductive structure **24**. A portion of the first conductive via **15** and a portion of the second connection via **28** may be embedded in the first bonding layer **51**. The first electrical path (including the first vertical electrical path **71**) may pass through the first bonding layer **51**. In addition, the first bonding layer **51** may extend into the second opening **247** to contact the first portion **244a** of the second test pad **244**.

[0086] Similarly, the second conductive via **25** and the third connection via **38** may contact each other or may be fused together to form a second interconnection pillar **55**. The second interconnection pillar **55** may be a monolithic structure, and there may be no interface between the second conductive via **25** and the third connection via **38**. The second interconnection pillar **55** may electrically connecting the second conductive structure **24** and the third conductive structure **34**, and may form the second electrical path (including the second vertical electrical path **72**). In addition, the second bottom passivation layer **26** and the third top passivation layer **37** may contact each other or may be fused together to form a second bonding layer **52**. The second bonding layer **52** may be a monolithic structure, and there may be no interface between the second bottom passivation layer **26** and the third top passivation layer **37**. The second bonding layer **52** may be used for bonding the second substrate **20** and the third conductive structure **34**. A portion of the second conductive via **25** and a portion of the third connection via **38** may be embedded in the second bonding layer **52**. The second electrical path (including the second vertical electrical path **72**) may pass through the second bonding layer **52**. In addition, the second bonding layer **52** may extend into the third opening **347** to contact the first portion **344a** of the third test pad **344**.

[0087] Similarly, the third conductive via **35** and the fourth connection via **48** may contact each other or may be fused together to form a third interconnection pillar **56**. The third interconnection pillar **56** may be a monolithic structure, and there may be no interface between the third conductive via **35** and the fourth connection via **48**. The third interconnection pillar **56** may electrically connecting the third conductive structure **34** and the fourth conductive structure **44**, and may form the third electrical path (including the third vertical electrical path **73**). In addition, the third bottom passivation layer **36** and the fourth top passivation layer **47** may contact each other or may be fused together to form a third bonding layer **53**. The third bonding layer **53** may be a monolithic structure, and there may be no interface between the third bottom passivation layer **36** and the fourth top passivation layer **47**. The third bonding layer **53** may be used for bonding the third substrate **30** and the fourth conductive structure **44**. A portion of the third conductive via **35** and a portion of the fourth connection via **48** may be embedded in the third bonding layer **53**. The third electrical path (including the third vertical electrical path **73**) may pass through the third bonding layer **53**. In addition, the third bonding layer **53** may extend into the fourth opening **447** to contact

the first portion **444a** of the fourth test pad **444**.

[0088] FIG. **8** is an enlarged cross-sectional view of the first portion of the electronic structure **5a** of FIG. **7**. The first portion may be substantially a first electronic device **1**. The first electronic device **1** of FIG. **8** may be similar to the first electronic device **1** of FIG. **2**, except that the second surface **12** of the first electronic device **1**, the second surface **162** of the first bottom passivation layer **16** and the second surface **152** of the first conductive via **15** may be imaginary surfaces rather than actual surfaces.

[0089] FIG. **9** is an enlarged cross-sectional view of the second portion of the electronic structure **5a** of FIG. **7**. The second portion may be a substantially second electronic device **2**. The second electronic device **2** of FIG. **9** may be similar to the second electronic device **2** of FIG. **3**, except that the first surface **21** of the second electronic device **2**, the first surface **271** of the second top passivation layer **27** and the first surface **281** of the second connection via **28** may be imaginary surfaces rather than actual surfaces. In addition, the second surface **22** of the second electronic device **2**, the second surface **262** of the second bottom passivation layer **26** and the second surface **252** of the second conductive via **25** may be imaginary surfaces rather than actual surfaces.

[0090] FIG. **10** is an enlarged cross-sectional view of the third portion of the electronic structure **5a** of FIG. **7**. The third portion may be a substantially third electronic device **3**. The third electronic device **3** of FIG. **10** may be similar to the third electronic device **3** of FIG. **4**, except that the first surface **31** of the third electronic device **3**, the first surface **371** of the third top passivation layer **37** and the first surface **381** of the third connection via **38** may be imaginary surfaces rather than actual surfaces. In addition, the second surface **32** of the third electronic device **3**, the second surface **362** of the third bottom passivation layer **36** and the second surface **352** of the third conductive via **35** may be imaginary surfaces rather than actual surfaces.

[0091] FIG. **11** is an enlarged cross-sectional view of the fourth portion of the electronic structure **5a** of FIG. **7**. The fourth portion may be a substantially fourth electronic device **4**. The fourth electronic device **4** of FIG. **11** may be similar to the fourth electronic device **4** of FIG. **5**, except that the first surface **41** of the fourth electronic device **4**, the first surface **471** of the fourth top passivation layer **47** and the first surface **481** of the fourth connection via **48** may be imaginary surfaces rather than actual surfaces.

[0092] FIG. **12** is a schematic cross-sectional view of an electronic structure **5b** in accordance with some embodiments of the present disclosure. FIG. **13** is a partially enlarged top view of the electronic structure **5b** of FIG. **12**. The electronic structure **5b** of FIG. **12** may be similar to the electronic structure **5a** of FIG. **7**, and the differences are described as follows.

[0093] The first interconnection pillar **54b** may include one first conductive via **15** and a plurality of second connection vias **28**. A width of each of the second connection vias **28** (e.g., the width **W21** of the second connection via **28** of FIG. **9**) may be less than a width of the first conductive via **15** (e.g., the width **W12** of the first conductive via **15** of FIG. **8**). Further, the second interconnection pillar **55b** may include one second conductive via **25** and a plurality of third connection vias **38**. A width of each of the third connection vias **38** may be less than a width of the second conductive via **25**. In addition, the third interconnection pillar **56b** may include one third conductive via **35** and a plurality of fourth connection vias **48**. A width of each of the fourth connection vias **48** may be less than a width of the third conductive via **35**.

[0094] FIG. **14** is a schematic cross-sectional view of an electronic structure **5c** in accordance with some embodiments of the present disclosure. FIG. **15** is a partially enlarged top view of the electronic structure **5c** of FIG. **14**. The electronic structure **5c** of FIG. **14** may be similar to the electronic structure **5a** of FIG. **7**, except the positions of the interconnection pillars **54**, **55**, **56**. For example, a vertical portion of the first conductive via **15** may be disposed outside the vertical projection of the first test pad **144**. However, the vertical projection of the first conductive via **15** may still overlap the vertical projection of the first test pad **144**.

[0095] FIG. **16** to FIG. **25** illustrate various stages of a method of manufacturing an electronic

structure **5**, in accordance with some embodiments of the present disclosure.

[0096] Referring to FIG. **16**, a first substrate **10** may be provided. The first substrate **10** of FIG. **16** may be same as or similar to the first substrate **10** of FIG. **1** and FIG. **2**. The first substrate **10** may have a first surface **101** (e.g., a top surface) and a second surface **102** (e.g., a bottom surface) opposite to the first surface **101**. Then, a first conductive via **15** may be formed in the first substrate **10**, and a first conductive structure **14** may be formed on the first surface **101** of the first substrate **10**. The first conductive via **15** and the first conductive structure **14** may be same as or similar to the first conductive via **15** and the first conductive structure **14** of FIG. **1** and FIG. **2**, respectively. The first conductive via **15** may be electrically connected to the first conductive structure **14**. The first conductive structure **14** defines a first opening **147** to expose a first portion **144a** of a first test pad **144** thereof. The first conductive via **15** is disposed under the first test pad **144**.

[0097] The first substrate **10** may have a plurality of singulation lines **19** crossed with each other to define a plurality of units **1'**. Each of the units **1'** may correspond to the first electronic device **1** of FIG. **2**. Then, a testing process may be conducted by using a probe **60**. The probe **60** is provided to contact and test the exposed first portion **144a** of the first test pad **144** so as to determine the electrical property of the unit **1'**. Meanwhile, a probe mark **148** may be recessed from a top surface of the first test pad **144**.

[0098] Referring to FIG. **17**, the first substrate **10** may be thinned to expose a bottom portion of the first conductive via **15**. Then, a first bottom passivation layer **16** may be formed or disposed on the second surface **102** (e.g., bottom surface) of the first substrate **10**. The first bottom passivation layer **16** may surround or cover the exposed bottom portion of the first conductive via **15**. Then, a grinding process (e.g., chemical mechanical polishing (CMP)) may be conducted to a second surface **162** of the first bottom passivation layer **16** such that a second surface **152** (e.g., a bottom surface) of the first conductive via **15** may be substantially aligned with the second surface **162** of the first bottom passivation layer **16**. Then, the first substrate **10** and the first conductive structure **14** may be singulated along the singulation lines **19** as to form a plurality of first electronic devices **1** as shown in FIG. **2**.

[0099] Referring to FIG. **18**, a second substrate **20** may be provided. The second substrate **20** of FIG. **18** may be same as or similar to the second substrate **20** of FIG. **1** and FIG. **3**. The second substrate **20** may have a first surface **201** (e.g., a top surface) and a second surface **202** (e.g., a bottom surface) opposite to the first surface **201**. Then, a second conductive via **25** may be formed in the second substrate **20**, and a second conductive structure **24** may be formed on the first surface **201** of the second substrate **20**. The second conductive via **25** and the second conductive structure **24** may be same as or similar to the second conductive via **25** and the second conductive structure **24** of FIG. **1** and FIG. **3**, respectively. The second conductive via **25** may be electrically connected to the second conductive structure **24**. The second conductive structure **24** defines a second opening **247** to expose a first portion **244a** of a second test pad **244** thereof. The second conductive via **25** is disposed under the second test pad **244**.

[0100] The second substrate **20** may have a plurality of singulation lines **29** crossed with each other to define a plurality of units **2'**. Each of the units **2'** may correspond to the second electronic device **2** of FIG. **3**. Then, a testing process may be conducted by using a probe **60**. The probe **60** is provided to contact and test the exposed first portion **244a** of the second test pad **244** so as to determine the electrical property of the unit **2'**. Meanwhile, a probe mark **248** may be recessed from a top surface of the second test pad **244**.

[0101] Referring to FIG. **19**, a second top passivation layer **27** may be formed or disposed on the first surface **241** of the second conductive structure **24**. The second top passivation layer **27** may cover the second opening **247**. Then, a second connection via **28** may be formed to extend through the second top passivation layer **27** and a portion **2461** of the dielectric structure **246** of the second conductive structure **24** on the second test pad **244** so as to electrically connect and physically connect to the second portion **244b** of the second test pad **244**. Then, a grinding process (e.g.,

chemical mechanical polishing (CMP)) may be conducted to a first surface **271** (e.g., top surface) of the second top passivation layer **27** such that a first surface **281** (e.g., a top surface) of the second connection via **28** may be substantially aligned with the first surface **271** of the second top passivation layer **27**.

[0102] Referring to FIG. **20**, the second substrate **20** may be thinned to expose a bottom portion of the second conductive via **25**. Then, a second bottom passivation layer **26** may be formed or disposed on the second surface **202** (e.g., bottom surface) of the second substrate **20**. The second bottom passivation layer **26** may surround or cover the exposed bottom portion of the second conductive via **25**. Then, a grinding process (e.g., chemical mechanical polishing (CMP)) may be conducted to a second surface **262** of the second bottom passivation layer **26** such that a second surface **252** (e.g., a bottom surface) of the second conductive via **25** may be substantially aligned with the second surface **262** of the second bottom passivation layer **26**. Then, the second substrate **20** and the second conductive structure **24** may be singulated along the singulation lines **29** as to form a plurality of second electronic devices **2** as shown in FIG. **3**.

[0103] Referring to FIG. **21**, a third substrate **30** may be provided. The third substrate **30** of FIG. **21** may be same as or similar to the third substrate **30** of FIG. **1** and FIG. **4**. Then, a third conductive via **35** may be formed in the third substrate **30**, and a third conductive structure **34** may be formed on the first surface **301** of the third substrate **30**. The third conductive via **35** and the third conductive structure **34** may be same as or similar to the third conductive via **35** and the third conductive structure **34** of FIG. **1** and FIG. **4**, respectively. The third conductive via **35** may be electrically connected to the third conductive structure **34**. The third conductive structure **34** defines a third opening **347** to expose a first portion **344a** of a third test pad **344** thereof. The third conductive via **35** is disposed under the third test pad **344**.

[0104] The third substrate **30** may have a plurality of singulation lines **39** crossed with each other to define a plurality of units **3'**. Each of the units **3'** may correspond to the third electronic device **3** of FIG. **4**. Then, a testing process may be conducted by using a probe **60**. The probe **60** is provided to contact and test the exposed first portion **344a** of the third test pad **344** so as to determine the electrical property of the unit **3'**. Meanwhile, a probe mark **348** may be recessed from a top surface of the third test pad **344**.

[0105] Referring to FIG. **22**, a third top passivation layer **37** may be formed or disposed on the first surface **341** of the third conductive structure **34**. The third top passivation layer **37** may cover the third opening **347**. Then, a third connection via **38** may be formed to extend through the third top passivation layer **37** and a portion of the dielectric structure **346** of the third conductive structure **34** on the third test pad **344** so as to electrically connect and physically connect to the second portion **344b** of the third test pad **344**. Then, a grinding process (e.g., chemical mechanical polishing (CMP)) may be conducted to a first surface **371** (e.g., top surface) of the third top passivation layer **37** such that a first surface **381** (e.g., a top surface) of the third connection via **38** may be substantially aligned with the first surface **371** of the third top passivation layer **37**.

[0106] Referring to FIG. **23**, the third substrate **30** may be thinned to expose a bottom portion of the third conductive via **35**. Then, a third bottom passivation layer **36** may be formed or disposed on the second surface **302** (e.g., bottom surface) of the third substrate **30**. The third bottom passivation layer **36** may surround or cover the exposed bottom portion of the third conductive via **35**. Then, a grinding process (e.g., chemical mechanical polishing (CMP)) may be conducted to a second surface **362** of the third bottom passivation layer **36** such that a second surface **352** (e.g., a bottom surface) of the third conductive via **35** may be substantially aligned with the second surface **362** of the third bottom passivation layer **36**. Then, the third substrate **30** and the third conductive structure **34** may be singulated along the singulation lines **39** as to form a plurality of third electronic devices **3** as shown in FIG. **4**.

[0107] Referring to FIG. **24**, a fourth substrate **40** may be provided. The fourth substrate **40** of FIG. **24** may be same as or similar to the third substrate **40** of FIG. **1** and FIG. **5**. Then, a fourth

conductive structure **44** may be formed on the first surface **401** of the fourth substrate **40**. The fourth conductive structure **44** may be same as or similar to the fourth conductive structure **44** of FIG. **1** and FIG. **5**. The fourth conductive structure **44** defines a fourth opening **447** to expose a first portion **444a** of a fourth test pad **444** thereof.

[0108] The fourth substrate **40** may have a plurality of singulation lines **49** crossed with each other to define a plurality of units **4'**. Each of the units **4'** may correspond to the fourth electronic device **4** of FIG. **5**. Then, a testing process may be conducted by using a probe **60**. The probe **60** is provided to contact and test the exposed first portion **444a** of the fourth test pad **444** so as to determine the electrical property of the unit **4'**. Meanwhile, a probe mark **448** may be recessed from a top surface of the fourth test pad **444**.

[0109] Referring to FIG. **25**, a fourth top passivation layer **47** may be formed or disposed on the first surface **441** of the fourth conductive structure **44**. The fourth top passivation layer **47** may cover the fourth opening **447**. Then, a fourth connection via **48** may be formed to extend through the fourth top passivation layer **47** and a portion of the dielectric structure **446** of the fourth conductive structure **44** on the fourth test pad **444** so as to electrically connect and physically connect to the second portion **444b** of the fourth test pad **444**. Then, a grinding process (e.g., chemical mechanical polishing (CMP)) may be conducted to a first surface **471** (e.g., top surface) of the fourth top passivation layer **47** such that a first surface **481** (e.g., a top surface) of the fourth connection via **48** may be substantially aligned with the first surface **471** of the fourth top passivation layer **47**.

[0110] Then, the fourth substrate **40** and the fourth conductive structure **44** may be singulated along the singulation lines **49** as to form a plurality of fourth electronic devices **4** as shown in FIG. **5**.

[0111] Then, the third electronic device **3** is stacked on or disposed on the fourth electronic device **4**, the second electronic device **2** is stacked on or disposed on the third electronic device **3**, and the first electronic device **1** is stacked on or disposed on the second electronic device **2**. Thus, the first substrate **10** is stacked on the second conductive structure **24**, the second substrate **20** is stacked on the third conductive structure **34**, and the third substrate **30** is stacked on the fourth conductive structure **34**. Thus, the first electronic device **1**, the second electronic device **2**, the third electronic device **3** and the fourth electronic device **4** are electrically connected to each other so as to form the electronic structure **5** of FIG. **1**.

[0112] In some embodiments, the first electronic device **1**, the second electronic device **2**, the third electronic device **3** and the fourth electronic device **4** are connected to each other through hybrid bonding so as to form the electronic structure **5a** of FIG. **7**. As shown in FIG. **7**, the first bottom passivation layer **16** and the second top passivation layer **27** may contact each other or may be fused together to form a first bonding layer **51**. The second bottom passivation layer **26** and the third top passivation layer **37** may contact each other or may be fused together to form a second bonding layer **52**. The third bottom passivation layer **36** and the fourth top passivation layer **47** may contact each other or may be fused together to form a third bonding layer **53**. The first conductive via **15** and the second connection via **28** may contact each other or may be fused together to form a first interconnection pillar **54**. The second conductive via **25** and the third connection via **38** may contact each other or may be fused together to form a second interconnection pillar **55**. The third conductive via **35** and the fourth connection via **48** may contact each other or may be fused together to form a third interconnection pillar **56**.

[0113] FIG. **26** illustrates a flow chart of a method **80** of manufacturing an electronic structure **5** in accordance with some embodiments of the present disclosure.

[0114] In some embodiments, the method **80** may include a step **S81**, forming a first conductive via in a first substrate and a first conductive structure on the first substrate, wherein the first conductive via is electrically connected to the first conductive structure, the first conductive structure defines a first opening to expose a first test pad thereof, and the first conductive via is disposed under the first test pad. For example, as shown in FIG. **16**, a first conductive via **15** may be formed in a first substrate **10**, and a first conductive structure **14** may be formed on the first substrate **10**. The first

conductive via **15** is electrically connected to the first conductive structure **14**. The first conductive structure **14** defines a first opening **147** to expose a first test pad **144** thereof. The first conductive via **15** is disposed under the first test pad **144**.

[0115] In some embodiments, the method **80** may include a step **S82**, testing the first test pad. For example, as shown in FIG. **16**, the first test pad **144** is tested by using a probe **60**.

[0116] In some embodiments, the method **80** may include a step **S83**, thinning the first substrate to expose the first conductive via. For example, as shown in FIG. **17**, the first substrate **10** is thinned to expose the first conductive via **15**.

[0117] One aspect of the present disclosure provides an electronic device including a substrate, a conductive structure and at least one external connector. The conductive structure is disposed on the substrate and includes a test pad configured to be contacted by a probe during a testing process. The external connector is electrically connected to the conductive structure and is exposed from a surface of the electronic device for an external electrical connection. A vertical projection of the at least one external connector overlaps a vertical projection of the test pad.

[0118] Another aspect of the present disclosure provides an electronic structure including a first substrate, a first conductive structure and a second conductive structure. The first conductive structure is disposed over the first substrate, and includes a first test pad configured to be contacted by a probe during a testing process. The second conductive structure is disposed under the first substrate, and includes a second test pad configured to be contacted by a probe during a testing process. An electrical path between the first conductive structure and the second conductive structure is located between the first test pad and the second test pad.

[0119] Another aspect of the present disclosure provides a manufacturing method. The manufacturing method includes forming a first conductive via in a first substrate and a first conductive structure on the first substrate, wherein the first conductive via is electrically connected to the first conductive structure, the first conductive structure defines a first opening to expose a first test pad thereof, and the first conductive via is disposed under the first test pad; testing the first test pad; and thinning the first substrate to expose the first conductive via.

[0120] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

[0121] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. A method for manufacturing an electronic device, comprising: forming a first conductive via in a first substrate and a first conductive structure on the first substrate, wherein the first conductive via is electrically connected to the first conductive structure, the first conductive structure defines a first opening to expose a first test pad thereof, and the first conductive via is formed under the first test pad; testing the first test pad; and thinning the first substrate to expose the first conductive via.
2. The method of claim 1, further comprising: forming a first bottom passivation layer on a bottom

surface of the first substrate to surround the first conductive via; forming a second conductive structure on a second substrate, wherein the second conductive structure defines a second opening to expose a first portion of a second test pad thereof; testing the second test pad; forming a second connection via connected to a second portion of the second test pad; and stacking the first substrate on the second conductive structure, wherein the first conductive via is connected to the second connection via.

3. The method of claim 2, further comprising: forming a second top passivation layer on the second conductive structure, wherein the second top passivation layer covers the second opening, and the second connection via extends through the second top passivation layer.

4. The method of claim 3, further comprising: forming a first bottom passivation layer on a bottom surface of the first substrate to surround the first conductive via, wherein after stacking the first substrate on the second conductive structure, the first bottom passivation layer and the second top passivation layer are fused together to form a bonding layer bonding the first substrate and the second conductive structure.

5. A method for manufacturing an electronic structure, comprising: providing a first substrate; forming a first conductive structure disposed over the first substrate and including a first test pad configured to be contacted by a probe during a testing process; and forming a second conductive structure under the first substrate and including a second test pad configured to be contacted by a probe during a testing process, wherein an electrical path between the first conductive structure and the second conductive structure is located between the first test pad and the second test pad.

6. The method of claim 5, wherein the electrical path includes a vertical electrical path, and a projection of the vertical electrical path on the second test pad is within a projection of the first test pad on the second test pad; the vertical electrical path passes through the first substrate.

7. The method of claim 5, further comprising: forming an interconnection pillar electrically connecting the first conductive structure and the second conductive structure, wherein the interconnection pillar forms the electrical path, wherein the interconnection pillar includes a conductive via and a connection via, and a width of the connection via is less than a width of the conductive via, and the conductive via extends through the first substrate, and the connection via connects to the second test pad.

8. The method of claim 3, wherein the interconnection pillar includes a conductive via and a plurality of connection vias, and a width of each of the connection vias is less than a width of the conductive via.

9. The method of claim 5, further comprising: forming a bonding layer bonding the first substrate and the second conductive structure, and the electrical path passes through the bonding layer, wherein the second conductive structure defines an opening to expose a portion of the second test pad, and the bonding layer extends into the opening to contact the portion of the second test pad.
